



Loops: Code That Repeats

Ontario Math C3.1 — repeating events

Name:

Date:

★ I can use a repeat loop to make Bit do something many times.

What is a loop?

Think about a song with a chorus you sing again and again, or doing 5 jumping jacks in gym. Lots of things in real life repeat. Code can repeat too — with a loop.

A loop is a block that repeats. You put the actions inside it one time, and the loop runs them again and again for you — so you do not add the same block over and over. Bit will show you how.

Look at this loop

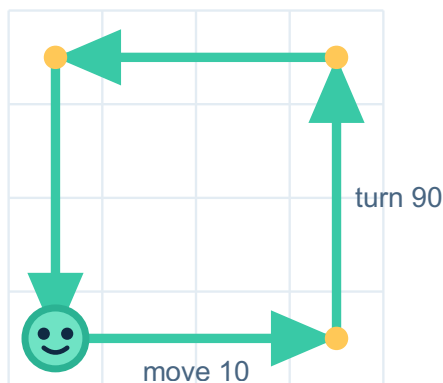
when green flag clicked

repeat 4

move 10 steps

turn 90 degrees

The 2 blocks inside repeat run 4 times. Bit moves and turns, 4 times. That draws a square.



Bit starts here

Bit repeats move and turn 4 times — and draws a square.

1 Count it

Look at the loop above. It says repeat 4. How many times does Bit MOVE? _____. How many times does Bit TURN? _____

2 Predict, then check

Now change repeat 4 to repeat 6. PREDICT first: how many times will Bit move? Write your guess. Then CHECK by tapping the inside blocks once for each repeat. Was your prediction right? Is it MORE or FEWER than before?

3 Write your own

Bit wants to JUMP 5 times. Write the number in the repeat block so the loop runs 5 times: repeat _____ → jump

4 Challenge

This time there are TWO blocks inside: repeat 3 → (jump, jump). Bit jumps twice each time the loop runs. How many jumps in total? Show how you worked it out.

Big idea

The number in the repeat block tells Bit how many times to run the blocks inside. A bigger number means more repeats — and if there is more than one block inside, every block runs each time around.